

Causal short-horizon flow-depth prediction at the USGS flume

The source provides high-frequency analog-laser measurements from five controlled gate releases, together with additional rangefinder data and the authors’ swath-processing script. This task keeps the analog depth measurements at the 33 m station in their native units and predicts the signal 200 ms ahead. It uses causal recent history, without future-peak alignment or target smoothing.

Five mechanistic or measurement hypotheses were tested after the baseline controls. The development-selected oscillatory memory model achieves 12.228 mm reserved MAE, beating the flexible control (13.688 mm) but losing to simple native persistence (11.667 mm). It therefore does not earn a positive claim of improved general prediction. The clearest supported finding is negative: unattenuated slope extrapolation roughly doubles error relative to persistence.

The oscillatory equation $\hat{h}=h+v \sin(\omega H)/\omega+a[1-\cos(\omega H)]/\omega^2$ propagates a local level, slope and curvature over $H=0.2$ s. Its fitted ω is an effective forecast parameter, not an independently measured flow-mode frequency. This distinction prevents a useful equation comparison from becoming a false constitutive-law claim.

Experimental scope and evaluation

Predict analog-laser depth 200 ms ahead, issuing every 50 ms after 400 ms of history. All features end at the issuance time; the target remains one native 1 kHz sample.

Calibration: Causal recent laser history is permitted at every issuance. No peak alignment, future smoothing or held-run fitted offset.

Validation unit: Complete gate-release run/date; first four columns develop, final 11 April 2024 column confirms. Five controlled events are not a field-hazard population.

Report individual complete-run error; one confirmation run supports no population confidence interval.

Target: analog laser flow-depth signal 200 ms ahead (m). Errors use m. The selected reference is **oscillatory_memory**. Selection was frozen before confirmation; diagnostic winners are not substituted afterward.

Predictor	Unit
h0	m
h	m; mean of last 20 native samples
v	m/s; causal 100 ms difference
a	m/s²; causal 100 ms second difference
median	m; median of last 20 native samples

Matched comparison

Model	Development MAE	Reserved MAE	Reserved worst-group MAE
constant_velocity	0.0289558	0.0231933	0.0231933
flexible	0.0142275	0.0136883	0.0136883
persistence	0.0137181	0.0116672	0.0116672
damped_velocity	0.013819	0.0120813	0.0120813
oscillatory_memory	0.0132228	0.0122284	0.0122284
depth_dependent_memory	0.0138252	0.0120793	0.0120793
front_recession_asymmetry	0.0138483	0.0120813	0.0120813
robust_surface_level	0.0139806	0.0121693	0.0121693

The primary error is mean group MAE. RMSE and signed bias are complementary; rows within a group do not establish independent replication. No accuracy percentage or industrial tolerance is invented.

Selected equation and coefficients

$$\hat{h}(t + \Delta) = h + \frac{v\sin(\omega\Delta)}{\omega} + \frac{a[1 - \cos(\omega\Delta)]}{\omega^2}$$

The horizon Delta is 0.2 s. Height h is in m, v in m/s, acceleration a in m/s^2 and omega in s^-1. This candidate failed its primary transfer comparison.

Coefficient	Frozen value
omega	24.3552821

All comparator expressions, numerical guards and full-precision values remain in EQUATIONS.md, rules.json and run.py.

Applicability and limitations

- This is a within-station nowcast, not advance warning upstream or a forecast before sensor detection.
- The archived swath-processing example trims around a future peak; that operation is excluded. Cross-instrument alignment is not inferred.
- Four debris flows and one water-only flood are described by the source, but file columns do not provide a reliable event-type map; no material-specific claim is made.

Only first 9 ms near-zero rows were seen in schema audit. Target horizon starts 600 ms; reserved future values were not inspected before freeze. All final outcomes become exposed.

Reproduction

Run python run.py in this final package to verify hashes and reproduce every saved prediction. Use python run.py --inputs new.csv --output predictions.csv --model reference only with the declared inputs and units. Numerical prediction evaluation does not certify novelty or mechanism. Source anchors and reserved observations are separate from predictor inputs.

Source: <https://doi.org/10.5066/P1HXHGZN>. See source/units audit and prior-art records in research history.